Adding HDFE Controls to did_multiplegt_dyn

Understanding the current controls() flow and a plan to extend it

Reading: did_multiplegt_dyn.ado (5,499 lines, January 17th 2026 version)

Companion: de Chaisemartin and D'Haultfoeuille (2024), Web Appendix 1.2

Deliverable

A concrete syntax + implementation plan for an absorb()-style HDFE option, with the variance implications.

Where controls live in the syntax

Entry point: line 53 of did_multiplegt_dyn.ado

```
syntax varlist(min=4 max=4 numeric) [if] [in] [, ...
    controls(varlist numeric)
    trends_nonparam(varlist numeric)
    trends_lin
    continuous(integer 0)
    weight(varlist numeric max=1)
    cluster(varlist numeric max=1)
    ... ]
```

KEY FACTS ABOUT CONTROLS() TODAY

- Numeric varlist only - factor variables (i.industry) are NOT accepted at the syntax level.
- Missing values in any control are dropped before estimation (line 282-286).
- Controls are first-differenced internally (Sec.3 of program), then residualized on time FEs.
- Time FEs are the ONLY high-dim object absorbed; trends_nonparam can interact them.
- To control for a time-invariant covariate, the help file tells the user to feed X*t themselves.

What controls() actually computes

Web Appendix 1.2 of dCdH (2024) - and where each step lives in the .ado

STEP-BY-STEP ON THE FIRST DIFFERENCES

- Take first differences: DY_g,t = Y_g,t - Y_g,t-1 and DX_g,t = X_g,t - X_g,t-1.
- Within each baseline-treatment cell d = D_g,1 and each trends_nonparam stratum, regress
- DY on DX + time FEs (control sample: not-yet-switchers, weighted by N_gt)
- to recover one slope vector theta_d per baseline level d.
- Residualize the long-difference outcome: Y_g,t - Y_g,t-l - (X_g,t - X_g,t-l) * theta_{D_g,1}.
- Plug the residualized long-diff into the standard event-study formula.
- Variance picks up an extra correction term U^var,X built from prod_X*_Ngt_XX and inv_Denom_d.

WHY THIS MATTERS FOR HDFE

Only ONE categorical dimension (time, optionally interacted with trends_nonparam) is absorbed inside this auxiliary regression. Adding more dimensions (industry-year, region-year, firm fixed effects in FD, etc.) is what HDFE buys us - and what users keep asking for.

Code map of the controls() block

did_multiplegt_dyn.ado - the four places that change for HDFE

Line	What	Why it matters for HDFE
L53	**syntax**	Where we add the new absorb() option.
L282-286	**sample selection**	Drops obs with missing controls; mirror for absorb() vars.
L331/336	**collapse**	Collapse to group*time keeps controls; absorb() vars must be carried.
L894-1098	**controls block**	First-differences + residualization + theta_d + Den^{-1}_d.
L985-991	**FD regression**	reg DY on DX + ibn.time (optionally x trends_nonparam). HDFE goes here.
L941	**resid_X*_time_FE**	Manual demeaning of DX by time-cell mean. Replace with absorb-aware demean.
L4582-4689	**long-diff block**	Residualizes the long-difference and builds U^var,X. Picks up new FEs.
L5145+	**placebo long-diff**	Same residualization for placebos - must stay in sync.

What's missing today: high-dim FE absorption

Concrete user requests the current option cannot satisfy

Things users ask for

- Industry x year FEs alongside controls
- Region x year FEs (e.g. state-by-year shocks)
- Firm or worker FEs on top of group + time
- Calendar-month or seasonal FEs that vary within (g,t)
- Categorical controls supplied as i.var without dummy expansion

Why workarounds break down

- Pre-residualizing Y and X by reghdfe before calling the command ignores the per-baseline theta_d structure.
- Manually creating dummies blows up #controls and triggers the 'singular Den_d' warning (line 1086).
- trends_nonparam helps only with time-invariant strata.
- Variance correction U^var,X assumes residualization by time FE only - adding FEs by hand silently mis-states SEs.

Proposed extension: absorb() option

Minimal-surface addition that plays nicely with everything already there

NEW SYNTAX

```
did_multiplegt_dyn Y G T D, effects(L) controls(X1 X2) ///
                   absorb(industry region#year) cluster(state)
```

SEMANTICS

- absorb() lists categorical variables (or interactions, a la reghdfe) to be partialled out of
- BOTH DY AND the DX_k's, in addition to time FEs, separately for each baseline-treatment cell d.
- Time FE is still absorbed automatically - users do not list it.
- If absorb() is empty, behavior is bit-identical to the current command (regression test).
- Categorical controls(i.var) get auto-routed to absorb() under the hood - kills the dummy blowup.

WHY AN OPTION, NOT A REWRITE

Keep the existing residualization machinery for users without HDFE needs (most of them). Add a single branch in the controls block that swaps the manual demeaning + reg call for an hdfe-backed equivalent. No change to the public estimator semantics when absorb() is empty.

Implementation plan - point estimates

Three surgical changes in the controls() block (L894-1098)

1. REPLACE THE TIME-FE DEMEANING OF DX (L931-941)

```
* CURRENT: bys time_XX d_sq_XX trends_nonparam : gegen avg_diff_Xk = total(N*diff_Xk) ...
* NEW    : if absorb() is empty -> keep current code (fast path, zero regression risk)
*          else                 -> hdfe diff_Xk diff_y_XX [aw=N_gt_XX] ///
*                                        if ever_change_d_XX==0 & d_sq_int_XX==`l',
*                                        absorb(time_XX `absorb' `trends_nonparam') gen(resid_)
```

2. REPLACE THE FD REGRESSION FOR THETA_D (L985-991)

```
* CURRENT: reg diff_y_XX `controlsXX' ibn.time_XX [aw=N_gt_XX] ///
*               if d_sq_int_XX==`l' & time_XX<F_g_XX, noconst
* NEW    : reghdfe diff_y_XX `controlsXX' [aw=N_gt_XX] ///
*               if d_sq_int_XX==`l' & time_XX<F_g_XX, ///
*               absorb(time_XX `absorb' `trends_nonparam') noconst
*          -> recover theta_d coefficients exactly as today (e(b))
```

3. LONG-DIFFERENCE RESIDUALIZATION (L4679, L5177)

diff_y_l - theta_d * diff_X_l stays UNCHANGED: HDFE only affects how theta_d is estimated, not how the long-difference is residualized. Same line, same coefficient slot - we just plug in the absorb-aware theta_d.

Implementation plan - variance (the tricky part)

U^var,X must change because residualization changes

CURRENT FORMULA (L941, L958, L4664)

```
resid_Xk_time_FE_XX = sqrt(N_gt) * (diff_Xk - avg_diff_Xk_by_time_cell)   // partial out time FE
prod_Xk_Ngt_XX      = resid_Xk_time_FE_XX * sqrt(N_gt)                     // enters U^var,X
inv_Denom_l_XX      = invsym( X'WX ) * sum(N_gt) * G                       // gradient block
```

WHAT CHANGES WITH ABSORB()

- resid_Xk_time_FE_XX must become residual w.r.t. (time FE) + absorb() + trends_nonparam.
- Same N_gt weighting - use _hdfe.dta-style two-step demeaning or a direct hdfe partialout call.
- X'WX in inv_Denom is built from MATRIX ACCUM on the new residuals (L1018) - no formula change,
- but degrees-of-freedom must subtract the absorbed FE dimensions for the singular-cell test (L1042).
- Companion-paper U^var,X correction (Sec.1.2) is derived under linear projection onto controls +
- FEs: enlarging the FE set is admissible. We owe a one-line proof sketch in the companion note.

DOF AND SINGULARITY BOOKKEEPING

The 'singular Den_d' branch (L1084-1090) already exists for collinear controls. With absorb(), we additionally need to know the count of absorbed FEs per baseline cell - reghdfe returns this as e(df_a). Subtract it from the # of usable obs in the singularity check, and add it to the error message so users know which dimension is starving the regression.

Interactions with existing options

Each existing option's behavior, and what absorb() must do alongside it

Option	What absorb() needs to do alongside it
`trends_nonparam(Z)`	Already interacts time FE with Z in residualization (L991). absorb() simply ADDS extra FEs - keep Z*time as today, plus the user's absorb terms.
`trends_lin`	Outcome is already a first-difference (L704); controls too (L716). absorb() applied on top works the same; no extra change needed.
`continuous(p)`	Adds polynomial-in-D_g,1 * cum-time-FE controls (L820-832). These are pre-residual controls in `controls' - HDFE residualizes them too. OK.
`by(W) / by_path()`	Estimation is re-run per subgroup. absorb() vars must be carried in the collapse() call (L331/336). One-line additive fix.
`predict_het(...)`	Already incompatible with controls() (L458-463). Same exclusion for absorb().
`bootstrap()`	controls_bs_orig_XX is replayed into the inner call (L2030/2033). Add absorb() to that local symmetrically.
`cluster(C)`	DOF adjustment for clustered SEs uses E_hat_denom (L4634-4641). Recompute with absorbed FE count subtracted - mirrors reghdfe convention.
`less_conservative_se`	Affects U^var,X scaling. Independent of absorb(); composes additively.

Validation plan

How we know we did not break anything

Regression tests (must pass)

1. absorb() empty -> bit-identical e(b), e(V), graphs vs. current master branch on:
   - the canonical staggered example in the help file
   - examples from prior_versions/* fixtures
   - controls() with 0, 1, 3 controls
2. controls() with i.industry vs. absorb(industry) on the same data -> identical theta_d, identical event-study estimates, narrower or equal SEs.
3. Bootstrap path (bootstrap option) replays absorb() correctly.

New behavior tests

4. Two-way absorb(industry year) on a panel where year already is the time variable - should warn 'year is the time FE, ignoring' (collinearity guard).
5. absorb(state#year) on simulated data with state-by-year shocks: bias drops to zero, SE grows as expected.
6. Singular-cell error (L1086) fires with the new FE dimension count and names the absorbed dimension.
7. cluster() x absorb() DOF matches reghdfe's e(df_r) convention on a side-by-side toy example.

Risks and open questions

Things I want your read on before coding

- Dependency: reghdfe / hdfe is not currently required by did_multiplegt_dyn (only gtools is, L65).
- -> add a soft check at L66-style block; offer ssc install hdfe if missing.
- Variance derivation: the companion paper's U^var,X is written for projection on time FE + X.
- Extending it to arbitrary absorb() needs a one-paragraph addendum. I will draft it for review.
- Performance: hdfe per baseline-treatment cell d can be slow if D_g,1 has many levels.
- Mitigation: fall back to direct demean when |absorb|=1 (current fast path).
- Scope creep: should absorb() also accept slope FEs (e.g. absorb(group##c.time))?
- My recommendation: NO for v1 - that overlaps with trends_lin and complicates the variance.
- Naming: absorb() matches reghdfe; alternatives are fe() or hdfe_controls(). My vote: absorb().

Suggested rollout

From plan to merged code

	What	Detail
Step 1	**Prototype on a fork**	Patch L894-1098 + L4582-4689 behind 'if `absorb' != ""' branches. Keep fast path untouched.
Step 2	**Smoke-test on help-file example**	Verify absorb() empty == master; absorb(i.var) == controls(i.var dummies).
Step 3	**Write companion-note addendum**	One-paragraph extension of Sec.1.2 covering arbitrary projection. Send for review.
Step 4	**Add fixtures to prior_versions/**	Lock in regression behavior before merging.
Step 5	**Document absorb() in .sthlp**	Mirror reghdfe wording; add an example block.
Step 6	**Ship behind a feature flag**	Surface as experimental for one release; promote to default in the next version.

Open invitation: I can start with Steps 1-2 on a worktree and come back with a working prototype + a side-by-side benchmark before touching the variance derivation. Tell me if you'd rather lock the math first.